

Project Outline and Expectations

ADSC2030

Due date for written report:

April 4th (Last day of class)

Authors will select their own data for analysis. **The data must come from a designed experiment (multiple experiments are encouraged) that we have covered in class.** Simulated data are permitted. *Data will be different than the sources we have used in the examples.* Authors are encouraged to submit a four-sentence abstract identifying their data source(s) and hypothesis/hypotheses prior to Test I. *Authors are responsible for ensuring the project topics are of the required standards.* Projects must be unique from other groups and topics are chosen on a first-come first served basis. Project topics and data must significantly differ from the authors' ADSC2020 project topics. **Self-plagiarism is a direct violation of the TRU Academic Integrity Policy.** *Instructor's consent for the abstract does not constitute or waive a potential academic integrity violation.*

The minimum group size is 1, while the maximum group size is 3, no exception. Collaboration in groups of 2 or 3 is strongly encouraged, as group size will have no impact on the grading standards. *The instructor will not deal with group issues.*

Abstracts and reports should be submitted in Moodle with all group members clearly listed.

If you are having trouble finding data, please speak with the instructor and we can set up some simulations or try to find existing data.

Report Instructions:

- The report must include a title page that includes, the authors names and student numbers, and a table of contents (see the example R Markdown header for help with setting this up).
- Data description: give a detailed description of the data what kind of variables are included, which designs were used to collect the data, and where the data come from.
- The report should **clearly define the objective(s) of the experiment(s) and the response variable(s)**.
- You may conduct your own experiment(s) to collect data.
- This project is intended to be a natural demonstration of analysis of data collected through an experimental design. The project must demonstrate your abilities in the following areas:
 - Analysis of variance (ANOVA)
 - Analysis of the factors
 - Selection of the number of replicates/blocks
 - Checking model assumptions
 - Interpretation and explanation of results
- **The report will be no longer than 12 pages excluding the title page and any appendices.**
- Please add a short appendix giving a description roles each team member played in the project.
- Please **only** include code in the main body of the report if it is deemed to be extremely important. All other code should be placed in an appendix (see the link in the example R Markdown header for help with setting this up).
- You may use any software you feel comfortable using. Using R is strongly encouraged and all code will need to be submitted.
- **As always, the TRU academic integrity policy is in effect. You must cite all external resources.**

Suggested Report Structure

1 Introduction (after title page) (1 page)

- Briefly discuss data.
- Data description (tables can be placed in appendix).
- State the overall objective(s) of the experiment(s).

2 Research Objectives (2 pages)

- Clearly state the research objective(s) (officially).
- State the hypothesis/hypotheses.
- May include multiple research questions/hypotheses (state them all here).

3 Analysis Selection (2-3 pages)

- Which tools will you use to analyze data and why?
- *I suggest that you select data collected using two different designs.*
- *Non-essential graphics can go in the appendix.*

4 Results (4-5 pages)

- **Clearly** present results including effects estimates, hypothesis tests about the model(s), ANOVA tables, diagnostics, etc.
- Interpret the results (how do these results relate to the experiment(s)' objectives?)

5 Conclusions

- Draw the project together with some overall conclusions.

Appendices

- A. – Description of the roles that each team member played in the project.
- B. – Additional graphics, code, and results.